

# Grid-Aware Techno-Economic Optimization of Residential PV-BESS Systems Across German City Tiers

## IEEE-Style Project Report

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Domain: Renewable Energy Systems, PV-BESS Optimization, Grid Support Analytics

|                       |                                                                                                   |
|-----------------------|---------------------------------------------------------------------------------------------------|
| Project type          | Simulation-based techno-economic and environmental optimization                                   |
| Cities studied        | Berlin, Munich, Hamburg, Freiburg, Leipzig, Dresden, Oldenburg, Cottbus, Passau                   |
| System search space   | PV: 3, 5, 7, 10 kW; Battery: 0, 3, 5, 8, 10, 12 kWh; Loads: 3500, 4200, 5000 kWh/year             |
| Main outputs          | NPV, payback period, self-sufficiency, CO2 savings, grid import/export, peak grid reduction proxy |
| Total simulated cases | 648                                                                                               |

### ABSTRACT

This project evaluates residential photovoltaic-battery energy storage system (PV-BESS) configurations across Tier 1, Tier 2 and Tier 3 German cities. A Python-based simulation framework estimates annual PV generation, self-consumption, self-sufficiency, grid import, grid export, payback period, net present value (NPV), lifetime carbon dioxide savings, battery degradation and a peak grid reduction proxy. The optimization considers nine German cities, three household load levels, four PV sizes and six battery capacities, producing 648 techno-economic cases. Results show that mid-sized Tier 2 cities have the strongest average NPV, while Freiburg, Passau and Munich emerge as the best-performing city-level cases. The analysis also shows that electricity price escalation has the largest positive effect on NPV, whereas lower feed-in tariffs reduce profitability. The work demonstrates how simplified grid-aware PV-BESS optimization can support early-stage renewable energy planning, residential energy decision-making and admission-oriented renewable energy project portfolios.

Keywords - PV-BESS, rooftop solar, battery energy storage, net present value, payback period, self-sufficiency, German cities, techno-economic optimization, grid-aware renewable energy.

### I. INTRODUCTION

The German residential energy transition increasingly depends on distributed solar photovoltaic generation, behind-the-meter battery storage and smarter interaction with the distribution grid. Rooftop PV reduces grid electricity purchases, while battery storage shifts midday solar surplus toward evening household demand. However, the financial and environmental value of a PV-BESS installation depends strongly on local solar yield, household load, system size, capital cost, retail electricity price, feed-in tariff and grid-related value assumptions.

This project studies how residential PV-BESS performance varies across different German city tiers. The selected cities represent major metropolitan areas, mid-sized solar-relevant cities and smaller regional locations. The objective is not to produce a bankable engineering design, but to create a transparent simulation model that compares system configurations and identifies high-performing combinations using consistent assumptions.

### II. RESEARCH OBJECTIVE AND SCOPE

The central research question is: How does the techno-economic and grid-impact performance of residential PV-BESS systems vary across Tier 1, Tier 2 and Tier 3 German cities, and what system size provides the best balance between cost savings, self-sufficiency, CO2 reduction and grid support?

- Estimate annual PV generation for each city using city-specific PV yield assumptions.
- Evaluate PV and battery combinations over multiple household load levels.
- Rank configurations using NPV, payback period, self-sufficiency and CO2 savings.
- Compare average results across Tier 1, Tier 2 and Tier 3 city categories.
- Perform sensitivity analysis for electricity price increase, battery cost reduction and feed-in tariff reduction.

### III. SYSTEM MODEL AND METHODOLOGY

The simulation was implemented in Python using NumPy, Pandas and Matplotlib. For each city, the annual PV energy output was estimated as the product of installed PV size and annual PV yield in kWh/kW/year. Household load was varied across three annual demand levels, while PV capacity and battery capacity were varied across discrete design options. The model calculates direct self-consumption first, then increases usable PV share when a battery is present. The total self-consumption ratio is capped at 85 percent to avoid unrealistic utilization.

The financial model estimates annual savings from self-consumed PV energy, feed-in revenue from exported PV energy, annual maintenance cost and discounted lifetime benefit. NPV is calculated by subtracting initial system cost from the discounted lifetime net benefit. Payback period is calculated as the ratio of system cost to annual net benefit. A simplified LCOE proxy is also calculated by dividing capital cost by lifetime PV generation.

The environmental model estimates annual CO<sub>2</sub> savings by multiplying PV energy used by the household by the grid emission factor. Lifetime CO<sub>2</sub> savings are calculated over a 15-year project life. Battery degradation is represented as a fixed annual health reduction, and battery replacement year is identified when the battery state of health falls to the replacement threshold.

#### A. Main Equations

Annual PV generation:  $E_{PV} = P_{PV} \times Y_{city}$ , where  $P_{PV}$  is installed PV capacity in kW and  $Y_{city}$  is annual PV yield in kWh/kW/year.

Annual net benefit:  $B = E_{used} \times C_{grid} + E_{export} \times C_{feed-in} - C_{system} \times m$ , where  $E_{used}$  is PV energy consumed by the household,  $C_{grid}$  is electricity price,  $E_{export}$  is exported PV energy,  $C_{feed-in}$  is feed-in tariff and  $m$  is the maintenance rate.

Net present value:  $NPV = \sum \text{from year 1 to } N \text{ of } B/(1+r)^{\text{year}} - C_{system}$ , where  $r$  is the discount rate and  $N$  is project lifetime.

Lifetime CO<sub>2</sub> savings:  $CO_2_{life} = E_{used} \times EF_{grid} \times N / 1000$ , where  $EF_{grid}$  is the grid emission factor in kg CO<sub>2</sub>/kWh.

### IV. INPUT DATA AND ASSUMPTIONS

Table I. Key Simulation Assumptions

| Parameter              | Value                             |
|------------------------|-----------------------------------|
| PV sizes               | 3, 5, 7, 10 kW                    |
| Battery sizes          | 0, 3, 5, 8, 10, 12 kWh            |
| Household annual loads | 3500, 4200, 5000 kWh/year         |
| Electricity price      | 0.38 EUR/kWh                      |
| Feed-in tariff         | 0.08 EUR/kWh                      |
| PV cost                | 1300 EUR/kW                       |
| Battery cost           | 650 EUR/kWh                       |
| Maintenance rate       | 1 percent of system cost per year |
| Discount rate          | 4 percent                         |
| Project lifetime       | 15 years                          |
| Grid emission factor   | 0.38 kg CO <sub>2</sub> /kWh      |
| Battery degradation    | 2.5 percent per year              |
| Replacement threshold  | 70 percent battery health         |

Table II. City Tier Classification

| City Tier | Cities                     | Interpretation                                   |
|-----------|----------------------------|--------------------------------------------------|
| Tier 1    | Berlin, Munich, Hamburg    | Major metropolitan cities                        |
| Tier 2    | Freiburg, Leipzig, Dresden | Mid-sized cities with strong renewable relevance |
| Tier 3    | Oldenburg, Cottbus, Passau | Smaller/regional cities                          |

### V. RESULTS

The simulation generated 648 candidate configurations. Best systems were selected by sorting valid payback cases by NPV within each city and tier. The key result is that a 10 kW PV system paired with a small 3 kWh battery frequently produced the best NPV because it balances high PV generation with relatively moderate storage cost. Larger batteries improved self-sufficiency but often reduced NPV because the incremental battery cost exceeded the additional financial benefit.

Table III. Best System by City Tier

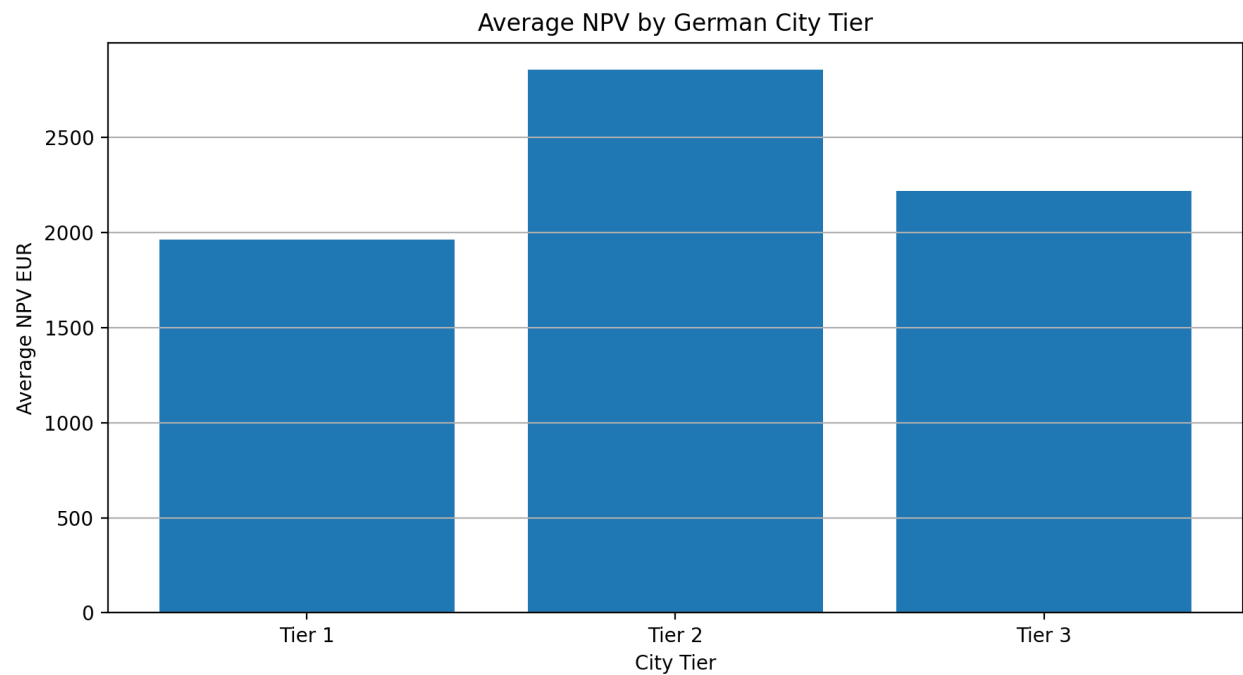
| Tier   | City     | Household Load kWh/year | PV Size kW | Battery Size kWh | NPV EUR   | Payback Years | Self Sufficiency Ratio % | Lifetime CO2 Savings tonnes |
|--------|----------|-------------------------|------------|------------------|-----------|---------------|--------------------------|-----------------------------|
| Tier 1 | Munich   | 5000                    | 10         | 3                | 9,188.02  | 6.890         | 98.70                    | 28.13                       |
| Tier 2 | Freiburg | 5000                    | 10         | 3                | 10,294.30 | 6.580         | 100.00                   | 28.50                       |
| Tier 3 | Passau   | 5000                    | 10         | 3                | 9,671.67  | 6.750         | 100.00                   | 28.50                       |

**Table IV. Best PV-BESS System by City**

| City      | Tier   | PV Size kW | Battery Size kWh | NPV EUR   | Payback Years | Self Sufficiency Ratio % | Lifetime CO2 Savings tonnes |
|-----------|--------|------------|------------------|-----------|---------------|--------------------------|-----------------------------|
| Freiburg  | Tier 2 | 10         | 3                | 10,294.30 | 6.580         | 100.00                   | 28.50                       |
| Passau    | Tier 3 | 10         | 3                | 9,671.67  | 6.750         | 100.00                   | 28.50                       |
| Munich    | Tier 1 | 10         | 3                | 9,188.02  | 6.890         | 98.70                    | 28.13                       |
| Cottbus   | Tier 3 | 10         | 3                | 8,696.59  | 7.030         | 96.82                    | 27.59                       |
| Dresden   | Tier 2 | 10         | 3                | 8,450.87  | 7.100         | 95.88                    | 27.33                       |
| Leipzig   | Tier 2 | 10         | 3                | 7,959.44  | 7.260         | 94.00                    | 26.79                       |
| Berlin    | Tier 1 | 10         | 3                | 7,468.00  | 7.410         | 92.12                    | 26.25                       |
| Hamburg   | Tier 1 | 10         | 5                | 6,459.31  | 7.960         | 99.00                    | 28.21                       |
| Oldenburg | Tier 3 | 10         | 5                | 6,186.91  | 8.050         | 97.90                    | 27.90                       |

**Table V. Tier-Level Average Performance Summary**

| Tier   | Annual PV Generation kWh | Self Sufficiency Ratio % | Payback Years | NPV EUR  | Lifetime CO2 Savings tonnes | Peak Grid Reduction Proxy kWh/year |
|--------|--------------------------|--------------------------|---------------|----------|-----------------------------|------------------------------------|
| Tier 1 | 6,104.17                 | 72.65                    | 9.871         | 1,961.31 | 17.32                       | 819.74                             |
| Tier 2 | 6,604.17                 | 75.79                    | 9.221         | 2,855.74 | 18.09                       | 708.28                             |
| Tier 3 | 6,250.00                 | 73.52                    | 9.699         | 2,216.92 | 17.53                       | 643.26                             |



*Fig. 1. Average NPV by German City Tier.*

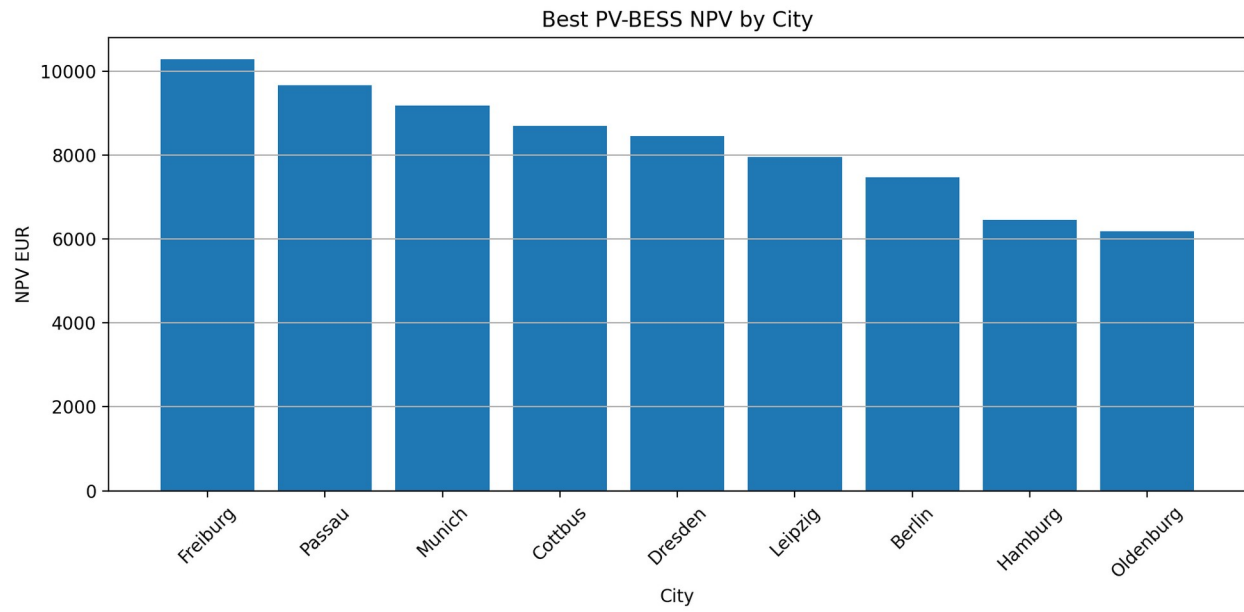


Fig. 2. Best PV-BESS NPV by City.

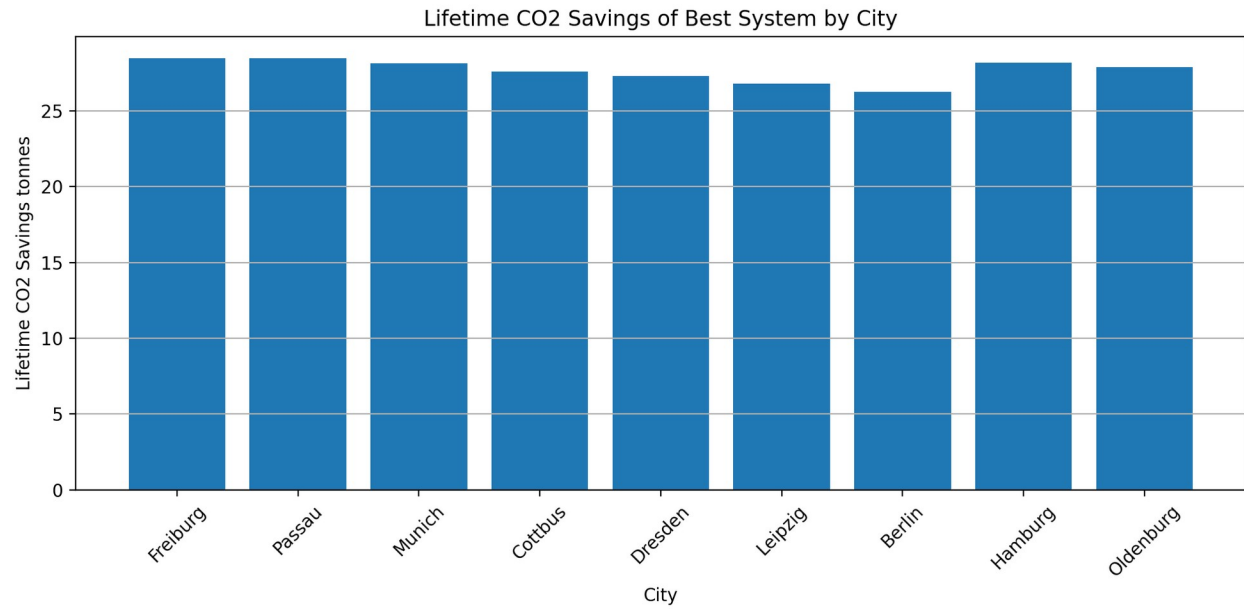


Fig. 3. Lifetime CO2 Savings of Best System by City.

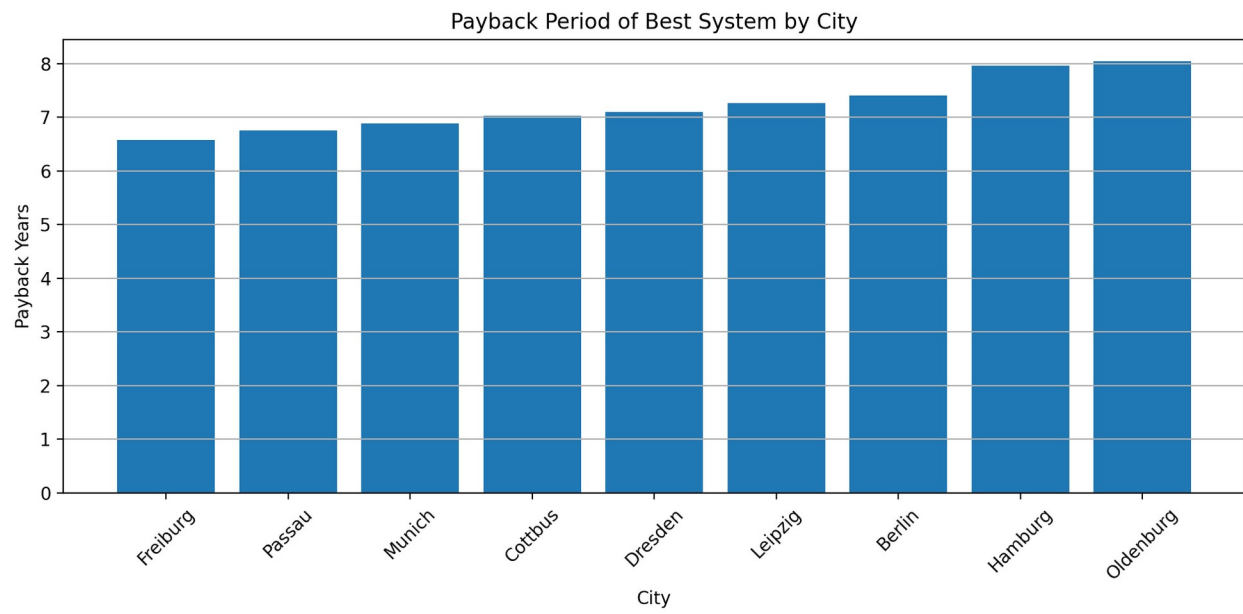


Fig. 4. Payback Period of Best System by City.

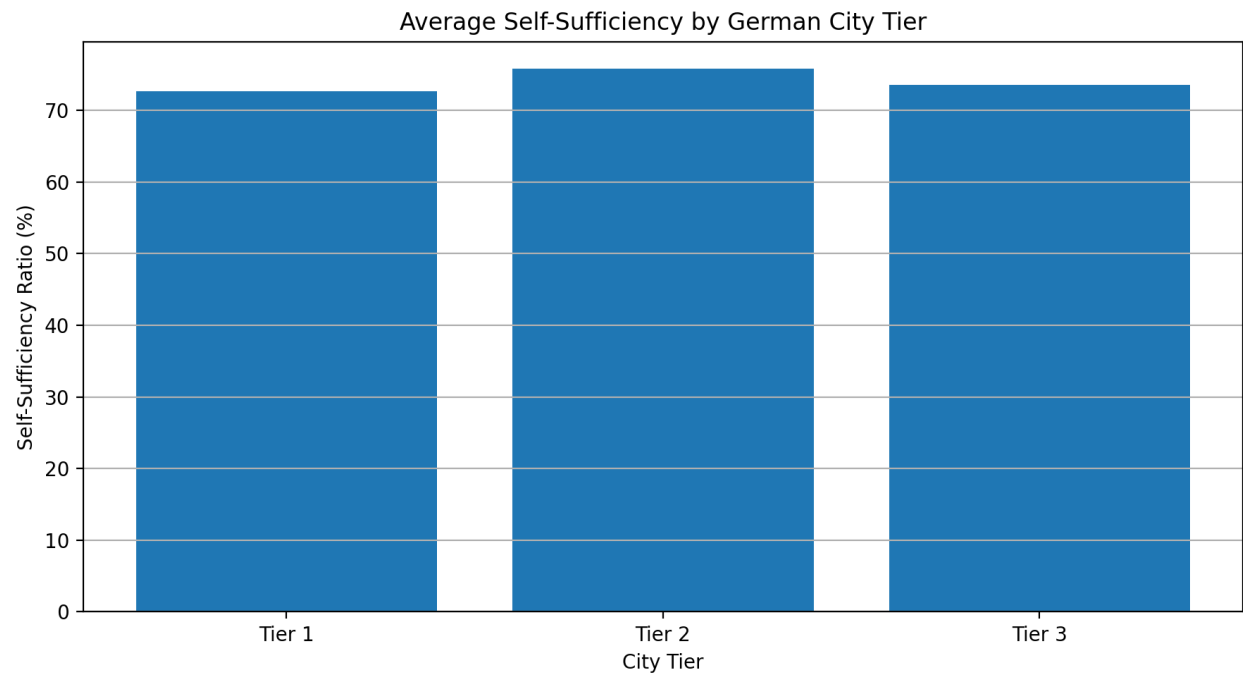


Fig. 5. Average Self-Sufficiency by German City Tier.

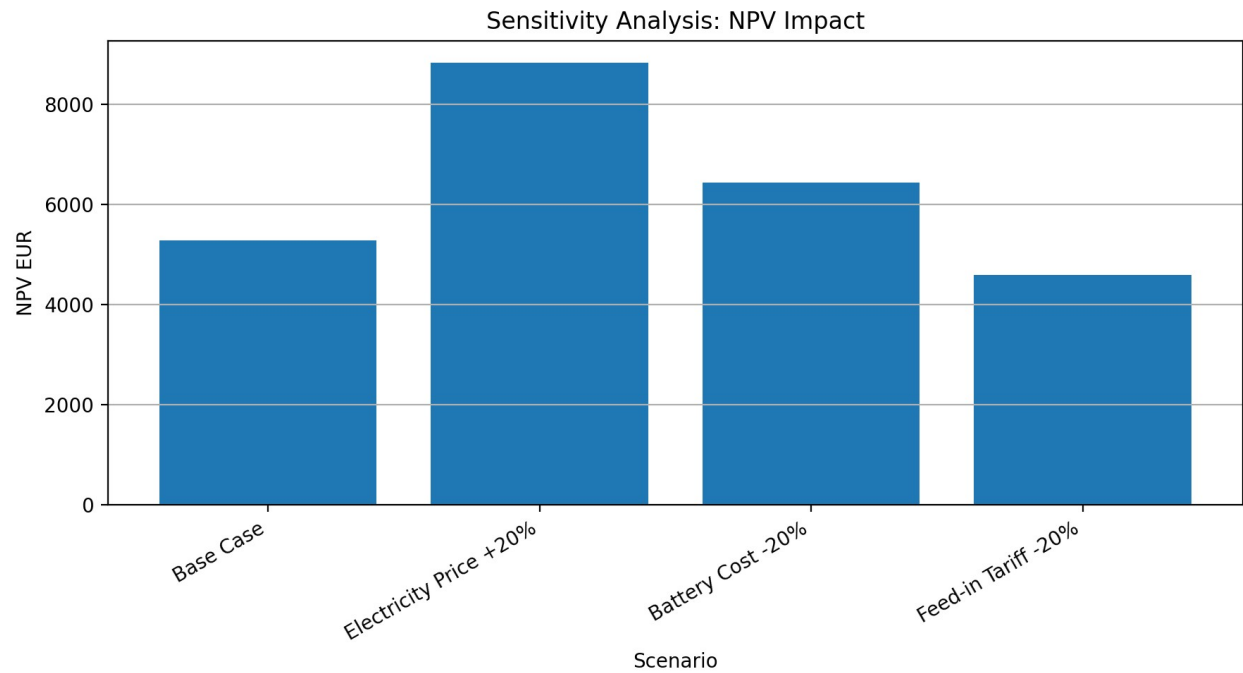


Fig. 6. Sensitivity Analysis: NPV Impact.

## VI. DISCUSSION

The tier comparison indicates that Tier 2 cities achieve the highest average NPV, approximately 2855.74 EUR, compared with 2216.92 EUR for Tier 3 and 1961.31 EUR for Tier 1. This result is consistent with the assumed higher PV yield in cities such as Freiburg and Dresden, combined with a moderate grid stress factor and strong generation potential. Tier 1 cities show relatively high grid stress benefit but lower average NPV in this simplified model because Hamburg and Berlin have lower assumed PV yields compared with the best southern or mid-sized cities.

Freiburg achieves the highest city-level NPV at 10294.30 EUR for the best system. Passau and Munich also show strong NPV performance. The best city-level cases generally use 10 kW PV capacity with 3 kWh storage, except Hamburg where a 5 kWh battery is selected. This indicates that small storage can increase self-consumption and grid support without imposing excessive capital cost.

The payback chart shows shorter payback for cities with stronger solar yield and higher self-consumed PV value. Freiburg has the shortest best-case payback at approximately 6.58 years, while Oldenburg has a longer payback of approximately 8.05 years. Despite similar lifetime CO<sub>2</sub> savings across cities, financial return varies more strongly because PV yield and export volume influence annual benefit and NPV.

The self-sufficiency tier chart shows that all city tiers can reach average self-sufficiency ratios above 70 percent under the simulated configuration set. Tier 2 again performs best, with average self-sufficiency near 75.79 percent. This reinforces the value of matching PV-BESS sizing to local solar resource and household consumption profile.

## VII. SENSITIVITY ANALYSIS

The sensitivity analysis was conducted using Freiburg, 7 kW PV and 8 kWh battery as a representative test case. Under the base case, the NPV is 5279.48 EUR and payback is 8.12 years. A 20 percent electricity price increase raises NPV to 8828.47 EUR and reduces payback to 6.87 years, making retail electricity price the most influential scenario among the tested cases. A 20 percent reduction in battery cost increases NPV to 6435.11 EUR. A 20 percent reduction in feed-in tariff decreases NPV to 4594.59 EUR.

Table VI. Sensitivity Analysis Results

| Case                   | City     | PV Size kW | Battery Size kWh | System Cost EUR | Annual Benefit EUR | Payback Years | NPV EUR  |
|------------------------|----------|------------|------------------|-----------------|--------------------|---------------|----------|
| Base Case              | Freiburg | 7          | 8                | 14,300.00       | 1,761.00           | 8.120         | 5,279.48 |
| Electricity Price +20% | Freiburg | 7          | 8                | 14,300.00       | 2,080.20           | 6.870         | 8,828.47 |
| Battery Cost -20%      | Freiburg | 7          | 8                | 13,260.00       | 1,771.40           | 7.490         | 6,435.11 |
| Feed-in Tariff -20%    | Freiburg | 7          | 8                | 14,300.00       | 1,699.40           | 8.410         | 4,594.59 |

## VIII. LIMITATIONS

This project is an academic and portfolio-oriented techno-economic simulation. It uses simplified annual PV yield assumptions instead of hourly PVGIS/weather-based generation profiles. Household demand is represented as annual load values rather than hourly load curves. Battery dispatch is represented through a self-consumption uplift approximation rather than chronological charge-discharge simulation. Grid stress is represented through a proxy instead of actual feeder constraints, voltage profiles or time-of-use network charges. For a publishable engineering version, the model should be extended using hourly PVGIS data, German standard load profiles, dynamic tariffs, inverter limits, battery degradation based on cycling depth and local distribution grid constraints.

## IX. RELEVANCE TO RENEWABLE ENERGY STUDIES IN GERMANY

The project aligns strongly with MSc-level renewable energy topics in Germany because it combines solar PV generation, battery storage economics, decentralized energy systems, grid interaction, emissions analysis and Python-based energy data analytics. It is especially relevant for programs covering renewable energy management, smart grids, distributed energy resources, energy storage systems and techno-economic optimization. The work can be further improved by integrating PVGIS data, German household load profiles, tariff scenarios, feed-in tariff sensitivity and battery degradation models.

## X. CONCLUSION

The project demonstrates that PV-BESS performance in Germany is highly dependent on solar yield, electricity price, battery cost and system sizing. Among the simulated cases, Tier 2 cities show the strongest average economic performance, and Freiburg provides the highest best-case NPV. Small battery capacities paired with larger PV systems often provide the best economic trade-off, while larger batteries improve self-sufficiency but may weaken NPV. The sensitivity analysis confirms that higher retail electricity prices substantially improve the business case for residential PV-BESS systems. Overall, this project provides a structured and extendable foundation for residential PV-BESS optimization and can be developed into a stronger academic portfolio project by adding hourly data and real tariff structures.

## REFERENCES

- [1] O. Arpit, Python simulation code: Grid-Aware Techno-Economic Optimization of Residential PV-BESS Systems Across German City Tiers, project file attached with this report.
- [2] O. Arpit, full\_optimization\_results.csv, best\_system\_by\_city.csv, best\_system\_by\_tier.csv, tier\_summary.csv and sensitivity\_analysis.csv, project outputs attached with this report.
- [3] Standard financial evaluation metrics: net present value, payback period and discounted cash-flow analysis, applied in the Python simulation framework.
- [4] Standard residential PV-BESS performance indicators: self-consumption ratio, self-sufficiency ratio, grid import/export and lifecycle CO<sub>2</sub> savings, applied in the Python simulation framework.

## APPENDIX A. ATTACHED OUTPUT FILES

The complete project output package includes the report, CSV result files and generated figures. The full optimization CSV contains all 648 simulated cases, while the smaller CSV files provide best-system and summary views used in the report.

**Table VII. Output File Inventory**

| File                          | Description                                                               |
|-------------------------------|---------------------------------------------------------------------------|
| full_optimization_results.csv | Complete simulation output containing all 648 cases.                      |
| best_system_by_city.csv       | Highest-NPV valid configuration for each city.                            |
| best_system_by_tier.csv       | Highest-NPV valid configuration for each city tier.                       |
| tier_summary.csv              | Average performance metrics by Tier 1, Tier 2 and Tier 3.                 |
| sensitivity_analysis.csv      | Sensitivity cases for electricity price, battery cost and feed-in tariff. |
| average_npv_by_tier.png       | Chart showing average NPV by city tier.                                   |
| best_npv_by_city.png          | Chart showing best NPV by city.                                           |
| co2_savings_by_city.png       | Chart showing lifetime CO <sub>2</sub> savings by city.                   |
| payback_by_city.png           | Chart showing best-system payback by city.                                |
| self_sufficiency_by_tier.png  | Chart showing average self-sufficiency by city tier.                      |
| sensitivity_npv.png           | Chart showing NPV sensitivity analysis.                                   |